

Supplementary material for Reinforcement learning with forecast-loss rewards for sensor scheduling under power constraints

This document provides implementation details for the Prediction-Driven Proximal Policy Optimization (PD-PPO) scheduler, baseline definitions, additional protocol diagnostics, and the 24-seed descriptive aggregate referenced by the anonymous manuscript. It introduces no additional training or model selection.

Terminology guide

The following terms are used consistently in the manuscript and this supplement.

Term	Meaning in this study
PD-PPO scheduler	Complete scheduler comprising input construction, the policy, feasibility check, and execution constraint.
Primary forecaster	Temporal convolutional network (TCN) trained on the forecaster-training partition.
Frozen primary forecaster	Primary forecaster after its parameters are fixed for policy training and offline evaluation.
Policy	Parameterized masked distribution π_θ over feasible proposed action indices.
Meteorological core channel	Mandatory logical channel active in every channel subset.
Optional channel	Any of the other five logical channels.
Other optional channels	Basic radiometer channel and shielded temperature–humidity channel.
Specialist channels	Infrared snow-surface-temperature channel, laser disdrometer channel, and FC4 snow mass-flux channel.
Channel subset	Set of requested logical channels at one time step.
Activation vector	Binary mathematical representation of a channel subset.
Proposed action index u_t	Categorical policy selection identifying one proposed candidate channel subset.
Feasible-action mask	Indicator over candidate actions after the feasibility check.
Minimum dwell-time constraint	Keeps the previously executed channel subset active until its minimum dwell time is met.
Sample-and-hold value	Last valid measurement retained when no new measurement is available.
Observation mask	Indicator that a variable received a new observation at the current time step.
Current condition label c_t	Simulator-derived current operating-condition label used for training-loss weighting and offline evaluation, not as a deployed policy input.
Future-condition supervision label c_t^{sup}	First non-calm label from t through $t + 16$, or calm if none occurs; used only for policy-training supervision.
Channel sampling age	Time since the channel’s last ready sampling opportunity; distinct from variable-level age of information (AoI).
Validation-selected fixed-schedule baseline	Fixed channel subset selected on the calibration/validation partition.
AoI-priority baseline	Feasible dynamic baseline prioritizing age of information.
Round-robin baseline	Baseline that uses a fixed cyclic schedule over candidate channel subsets.
Random-feasible baseline	Baseline sampled from the feasible action set.
Privileged one-step look-ahead reference using next-step test targets	Reference that selects the feasible action with the lowest next-step forecast loss on the test partition while omitting longer dynamic consequences from its objective.
Warning-rule baseline	Deterministic baseline driven by simulated warning signals.
Privileged true-label baseline	Baseline using the exact label sequence $c_t, \dots, c_{t+16}$ at each decision.
Double DQN training configuration	Training-configuration comparison using the same state representation, reward, action set, and constraints; it omits behavior cloning pretraining, continuing advantage-weighted behavior cloning, and the auxiliary future-condition classifier used by the complete PD-PPO configuration.
Macro-averaged normalized forecast loss	Equal-weight mean of the three event-type-specific losses after calibration/validation-based normalization.

S1. Recent-history features and policy encoding

The scheduler input contains ten deterministic recent-history features. Let $\bar{x}_{t,j}$ be the sample-and-hold value normalized by forecaster training statistics and clipped to $[-5, 5]$, let $\Delta_{16}\bar{x}_{t,j} = \bar{x}_{t,j} - \bar{x}_{t-15,j}$, and let $C_{16}(\mathcal{J})$ be the largest 16-step observation fraction among variables in $\mathcal{J}$. Three event-related

recent-history features are

$$b_t^{\text{part}} = \tanh(0.45\bar{v}_{p,t} + 0.25\bar{d}_{p,t} + 0.15\bar{f}_t), \quad (\text{S1})$$

$$b_t^{\text{flux}} = \tanh(0.55\bar{f}_t + 0.25\bar{v}_{\text{wind},t} + 0.10\Delta_{16}\bar{f}_t), \quad (\text{S2})$$

$$b_t^{\text{therm}} = \tanh(0.45(\bar{T}_{a,t} - \bar{T}_{s,t}) - 0.10\Delta_{16}\bar{T}_{s,t}). \quad (\text{S3})$$

Here v_{wind} , f , v_p , d_p , T_a , and T_s denote wind speed, snow mass flux, particle velocity, particle diameter, air temperature, and snow-surface temperature. The three indicators are concatenated with current normalized wind speed, snow mass flux, particle velocity, the air-surface temperature difference, and the particle, flux, and thermal coverage terms. Every component is computed from sample-and-hold values and observation masks; this information is available at decision time.

The policy uses the aggregate warning signal q_t^{max} to combine two observation-history encoders,

$$h_t = (1 - g_t)f_0(o_t) + g_tf_1(o_t), \quad g_t = \text{sigmoid}(\alpha q_t^{\text{max}} + \beta), \quad (\text{S4})$$

where α , β , and both encoders are learned. A candidate activation vector $a^{(u)}$ is represented by the sum of learned embeddings of its active channels. Its policy score is the scaled inner product between the subset embedding and h_t , plus a learned subset bias. The feasible-action mask excludes infeasible action indices before categorical sampling. The critic receives o_t and q_t^{max} through a separate multilayer perceptron.

S2. Forecaster and policy training protocol

The primary forecaster is fitted on 78 rollouts containing 53,274 time steps, all drawn from the forecaster-training partition. Its parameters are then frozen during policy training and evaluation. The full-observation baseline contributes 18 rollouts. A feasible fixed-schedule source and the round-robin, AoI-priority, and random-feasible baselines contribute six each, as does each of the six fixed channel subsets. Each rollout contains 683 time steps. For each schedule, three starts are event centered and three are sampled uniformly from the forecaster-training partition. At reset, all channels are off, the 20-step observation history is initialized to the mean of that partition, and its observation masks are zero. No calibration/validation or test observation enters the training mixture.

Each event-balanced test window is evaluated as a separate rollout. At its start, all channel runtimes are off, the previous activation vector and remaining required activation time are zero, the sample-and-hold values and 20-step history are initialized to the configured state mean, and observation masks are zero. The policy or minimum dwell-time wrapper state is also reset. No pre-roll context is supplied. Metrics concatenate the eight 512-step rollouts; action-transition diagnostics therefore include the seven boundaries between separately reset windows. Continuous full-partition evaluation instead uses one reset at time step 64,750 followed by one 5,242-step rollout.

Table S1: Main protocol and PD-PPO scheduler hyperparameters for the reported experiment.

Item	Value
Reported seeds	117–140: pilot/model-selection seeds 117–118; post-selection evaluation seeds 119–140
Simulated series length	70,000 hourly time steps per seed
Chronological partitions	[0,24500) forecaster training; [24500,59500) policy training; [59500,64750) calibration/validation (normalization-factor computation and fixed-baseline selection); [64750,70000) test
Forecast lookback / horizon	20 / 8 time steps; uniform horizon weights
Event-balanced test windows	8 windows selected from test truth to balance event types and emphasize transport periods, 512 time steps each; selection uses no policy losses
Full test-partition evaluation	All 5,242 scoreable test time steps; same trained policies, channel subsets, and normalization factors
Candidate action set	6 candidate channel subsets
Mandatory channel	Meteorological core channel
Steady-state / startup-peak power limits	0.75 / 0.95
Constraint activation in the reported candidate set	Maximum steady/startup demand 0.74/0.92; all six candidates feasible at every step; optional-channel cardinality and minimum dwell time are the active restrictions
Policy timesteps	200,000
PPO rollout / batch / update epochs	1024 / 128 / 8
Recent-history features	16-step normalized values, slopes, and observation coverage; 10 features
Policy/critic network	two 128-unit feature encoders combined by the aggregate warning signal; 32-dimensional embeddings of candidate channel subsets; separate warning-aware critic
Learning rate, discount, GAE	0.0003, 0.99, 0.95
Clip, entropy, value coef.	0.2, 0.0075, 0.5
Continuing advantage-weighted behavior cloning and auxiliary future-condition classifier weights	$\beta_{BC} = 2.5$; $\beta_{cond} = 5.0$
Condition-specific training weights	particle/flux/thermal = 1.55/0.95/1.65; calm = 1.0
Calibration/validation-based normalization factors	event-type-specific median over fixed channel subsets for particle/flux/thermal; default overall-loss median for calm
Future-condition supervision label	first non-calm condition label from t through $t + 16$, or calm if none occurs; training only
Minimum dwell time / normalized switching-cost weight / warm-up interruption-cost weight	6 time steps; 0.001; 1.0
Fixed-baseline selection metric	macro-averaged normalized forecast loss
Event-type mix	particle/flux/thermal = 0.40/0.20/0.40
Event coverage target	0.68
Statistical resampling	100,000 percentile-bootstrap samples over paired seed differences; seed 20,260,718
Behavior cloning pretraining	10,000 steps; 18 epochs; batch size 256

Partitions are fixed before test evaluation; fixed-baseline selection and normalization factors use only the calibration/validation partition.

Table S2: Scalar-objective PPO configurations, Double DQN training configuration, and primary-forecaster training protocol.

Item	Value
Scalar-objective PPO variants	Same PPO policy and critic architecture, scheduler input, action set, constraints, behavior cloning pretraining, continuing advantage-weighted behavior cloning, and auxiliary future-condition classifier; replace forecast loss with normalized target AoI or a bounded diagonal uncertainty proxy
Double DQN training configuration	Dueling 128-unit network; 200,000 steps; experience replay 100,000; learning starts 5,000; batch 128; 3-step return; target update 1,000; ϵ 1.0 to 0.05 over 20%; learning rate 10^{-4} ; no behavior cloning pretraining, continuing advantage-weighted behavior cloning term, or auxiliary future-condition classifier
Primary forecaster architecture	residual TCN, 3 levels, 64 channels, kernel size 3, dropout 0.05
Forecaster-training mixture	78 rollouts (53,274 time steps). The mixture contains 18 full-observation rollouts, six for each dynamic baseline or feasible fixed-schedule source, and six for each fixed channel subset.
Normalization	forecaster input/target scales fitted before policy updates; calibration/validation-based normalization factors computed before policy optimization
Checkpoint/baseline handling	final policy checkpoint saved after the prescribed PPO updates; fixed baseline selected by lowest calibration/validation macro-averaged loss
Computation	GPU policy optimization; the same frozen primary forecaster for all test metrics

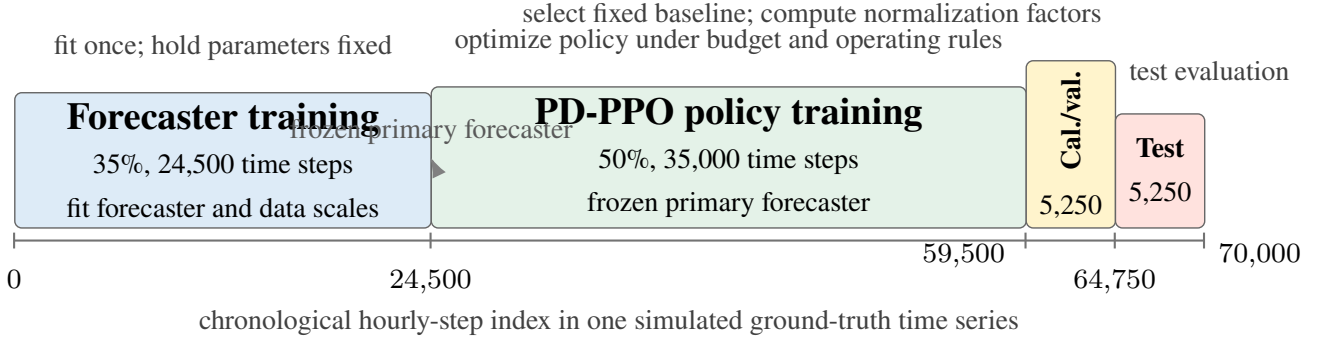


Figure S1: Chronological data partitions used in the experiments. Forecaster training, policy training, calibration/validation, and testing occupy separate intervals. The calibration/validation partition is used to select the fixed baseline and compute normalization factors and the supervised-action mapping before policy optimization; the test partition supplies the event-balanced test windows and evaluation over all scoreable test time steps. The two pilot/model-selection seeds use selected test-window metrics for the prespecified architecture comparison; the other 22 seeds are evaluated after that configuration is fixed.

S3. Matched scalar-objective PPO configurations

For the configured observed-variable set $\mathcal{J}$ and history length H , the AoI-objective PPO configuration uses

$$\mathcal{L}_{\text{AoI}}(t) = \frac{1}{|\mathcal{J}|} \sum_{j \in \mathcal{J}} \min \left(\frac{A_{t,j}^{\text{var}}}{H-1}, 1 \right), \quad (\text{S5})$$

where $A_{t,j}^{\text{var}}$ is the variable-level AoI, measured in time steps since the latest valid observation of variable j in the input history. A variable with no valid observation in the history receives unit loss.

The uncertainty-objective PPO configuration maintains a bounded diagonal predict-update variance proxy only for its reward. For each variable, it applies

$$P_{t,j}^- = \min(P_{t-1,j} + Q_j, P_{\max}), \quad P_{t,j} = [(P_{t,j}^-)^{-1} + R_{t,j}^{-1}]^{-1} \quad (\text{S6})$$

when a valid observation is available, and $P_{t,j} = P_{t,j}^-$ otherwise. $R_{t,j}$ is the inverse-variance fused measurement variance. Its bounded reward proxy is

$$\mathcal{L}_{\text{unc}}(t) = \frac{1}{|\mathcal{J}|} \sum_{j \in \mathcal{J}} \frac{P_{t,j}}{1 + P_{t,j}}. \quad (\text{S7})$$

Both scalar-objective PPO variants retain the same condition-specific training weights, behavior cloning pretraining, continuing advantage-weighted behavior cloning loss, and auxiliary future-condition classification loss as PD-PPO. The current operating-condition label c_t supplies the training weight. The future-condition supervision label c_t^{sup} is the first non-calm label from t through $t+16$, or calm if no event occurs in that window; it supplies the supervised target action and auxiliary future-condition classification target. Under the reported stride-one configuration both labeled-example indicators are one at every PPO step, so $\eta_t = \ell_t$; the general implementation constructs them separately. Neither label is available to the policy or critic at decision time. The diagonal uncertainty proxy is distinct from the general estimator covariance trace analyzed in Proposition 2 of the main manuscript.

S4. Physical platform motivation

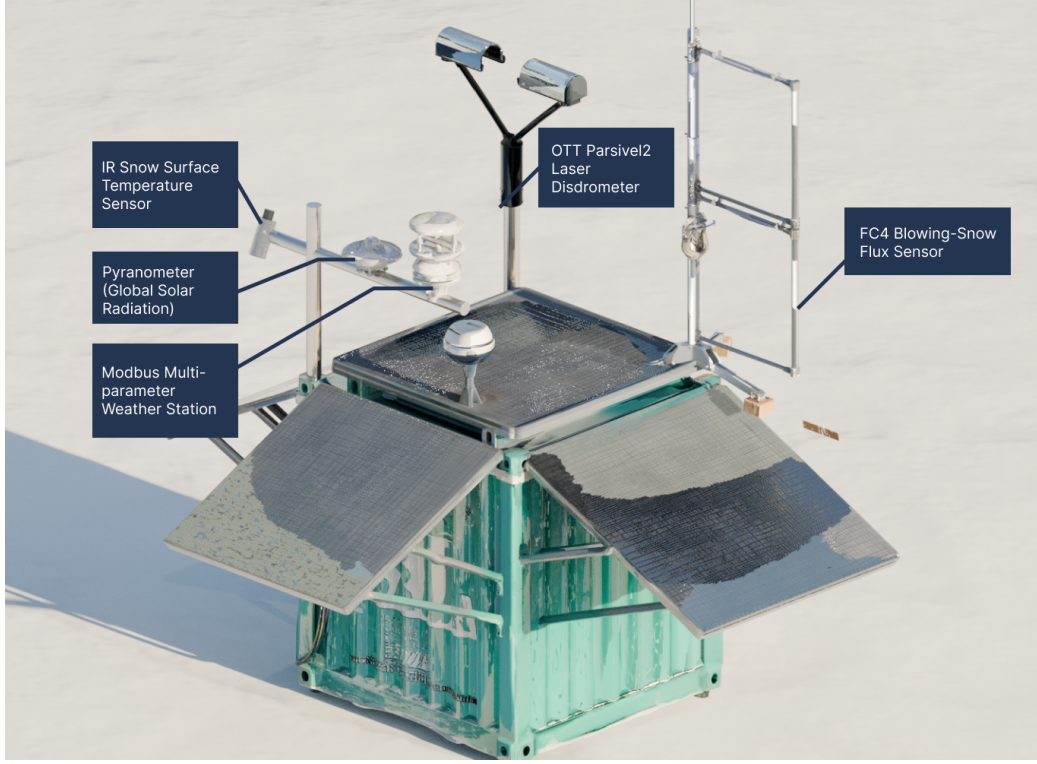


Figure S2: Rendered automatic weather station platform motivating the reported sensing abstraction. The logical channel table in the main manuscript defines the experimental sensing system.

S5. Simulation model and observation specification

The meteorological background series is generated by phase-randomizing contiguous reference time series from Antarctic AWS stations and matching the empirical marginal distribution of each variable. An alternating calm/storm semi-Markov process supplies longer calm and storm intervals. Their durations are sampled from a lognormal distribution, after which event pulses with bounded duration and gap are placed inside storm intervals. Wind credibility and hysteresis convert candidate pulses into event intervals. One of the particle, flux, or thermal event types is sampled for each complete event run.

Within an event run, each event-type process follows a first-order low-pass process

$$z_{t,c} = (1 - \alpha)z_{t-1,c} + \alpha\epsilon_{t,c}, \quad \epsilon_{t,c} \sim \mathcal{N}(0, 1), \quad (\text{S8})$$

and is standardized over its active interval. Event-type effects enter the target variables after a six-time-step lag. Table S3 lists the simulation-model parameters used in the reported experiment.

Table S4 records the target scales and task weights used by the primary and ridge forecasters. The current operating-condition label c_t selects a predeclared weight vector after an observation sequence has been produced by a scheduler; it never changes the scheduler input.

Table S3: Parameters of the simulation model. One time step equals one hour.

Component	Value	Role
Reference distributions from Antarctic AWS stations	Panda100, Panda200, Taishan	Phase-randomized Antarctic AWS series with empirical marginal matching.
Sequence length and interval	70,000; 1 h	One independently generated simulated ground-truth time series per seed.
Simulated blowing-snow event process	alternating semi-Markov storm/calm phases	Lognormal phase duration with log-scale standard deviation 0.35 and a minimum duration of 24 time steps.
Storm / within-storm event duty	0.65 / 0.78	Derived from the event-coverage target after clipping.
Median storm / calm duration	1,440 / 775 time steps	Parameters of the alternating calm/storm process.
Event pulse duration / gap	Uniform 28–80 / 8–10 time steps	Bursty blowing-snow episodes within storm phases.
Wind thresholds and event margin	8 / 12 / 1.4 m s ⁻¹	Weak threshold, strong threshold, and event wind-floor margin.
Hysteresis on / off	0.60 / 0.30	Converts candidate pulses and wind credibility into event intervals.
Particle / flux / thermal probability	0.40 / 0.20 / 0.40	One event type is sampled for each complete event run.
Event-type process smoothing	0.65	Low-pass coefficient for event-type-specific simulated processes.
Target-effect lag	6 time steps	Delay from the event-type process to target effects.
Particle process scales	0.62 mm; 8.2 m s ⁻¹	Diameter and particle-velocity effects.
Flux process map	scale 0.0014; offset 1.25; clip 2.4	Linear event-type effect before clipping.
Thermal process scale	6.4 °C	Snow-surface-temperature effect.
Warning-signal lead / noise	16 time steps / 0.01	Linear pre-event ramp plus Gaussian noise, clipped to [0, 1].

Table S4: Target scales and event-type-dependent weights used with the primary forecaster. Zero-weight variables remain forecaster inputs but do not enter the reported loss.

Forecast target	Scale	Calm/default	Particle	Flux	Thermal
Air temperature	5	0.10	0	0	0.22
Snow-surface temperature	5	0.26	0	0	85
Wind speed	5	0.08	0.08	0.04	0.08
Wind direction sine	1	0	0	0	0
Wind direction cosine	1	0	0	0	0
Solar radiation	100	0	0	0	0
Snow mass flux	10 ⁻⁴	4.5	0.5	7	0.2
Particle diameter	0.2	42	105	5	1
Particle velocity	5	42	105	5	1

For each event type c , the simulated warning signal is

$$q_{t,c} = \text{clip}_{[0,1]}(b_{t,c} + \varepsilon_{t,c}), \quad (\text{S9})$$

where $b_{t,c} = 1$ during the event interval, increases linearly during the 16 time steps before onset, and is zero otherwise; $\varepsilon_{t,c}$ is Gaussian noise with standard deviation 0.01. The aggregate warning signal is $\max_c q_{t,c}$. The PD-PPO scheduler receives these four simulated warning signals but neither c_t nor c_t^{sup} . This is a simulation assumption and has not been validated as a field detector.

An active and ready channel observes variable j with configured probability $p_{i,j,c}$ and adds independent Gaussian noise with standard deviation $\sigma_{i,j,c}$. Table S5 gives the event-dependent values of the three specialist channels. The meteorological core channel uses standard deviations 0.18 m s⁻¹ for wind speed, 1.8 degrees for wind direction, 0.18 °C for air temperature, 0.90 percentage points for relative humidity, and 14 Pa for pressure. The radiometer uses 8 W m⁻², while the shielded temperature–humidity channel uses 0.06 °C and 0.30 percentage points.

Table S5: Availability probabilities and Gaussian noise standard deviations for the three specialist logical channels. Entries within a cell follow the variable order shown in the second column. “Other event” means an active non-matching event type.

Specialist logical channel	Variables	Calm (p, σ)	Other event (p, σ)	Matching event type (p, σ)
Infrared snow-surface-temperature channel	snow-surface temperature; thermal process	(0.20, 1.40); (0.10, 1.20)	(0.25, 1.60); (0.18, 1.20)	(0.96, 0.07); (0.98, 0.03)
Laser disdrometer channel	diameter; particle velocity; particle process	(0.10, 1.20); (0.10, 3.00); (0.05, 1.50)	(0.18, 0.90); (0.18, 2.80); (0.12, 1.20)	(0.96, 0.06); (0.96, 0.16); (0.98, 0.03)
FC4 snow mass-flux channel	snow mass flux; flux process	(0.10, 5.0×10^{-6}); (0.05, 1.50)	(0.18, 4.0×10^{-6}); (0.12, 1.20)	(0.96, 2.5×10^{-7}); (0.98, 0.03)

Noise is expressed in the physical units of each variable. The process variables are simulation-model constructs used to make event-dependent channel value explicit; they are distinct from the condition labels.

S6. Executable baseline definitions

The fixed channel subset selected on the calibration/validation partition is

$$a_{\text{fixed}} = \arg \min_{a \in \mathcal{A}} (L_{\text{macro, val}}(a), L_{\text{val}}(a), \text{index}(a)), \quad (\text{S10})$$

with lexicographic ordering. This fixed channel subset is selected before test evaluation. The warning-rule baseline uses the same calibration/validation procedure separately for calm, particle, flux, and thermal windows. At time step t , it chooses

$$\hat{c}_t = \begin{cases} \arg \max_c q_{t,c}, & \max_c q_{t,c} \geq 0.5, \\ \text{calm}, & \text{otherwise,} \end{cases} \quad (\text{S11})$$

and requests the channel subset selected on the calibration/validation partition for $\hat{c}_t$.

The privileged one-step look-ahead reference using next-step test targets snapshots the complete environment, executes each candidate once, restores the snapshot, and minimizes

$$\left(L_{\text{step}}(t; a), \frac{1}{N} \|a - a_{t-1}\|_1, \text{index}(a) \right). \quad (\text{S12})$$

It uses next-step test targets that are unavailable to the deployed scheduler. The privileged true-label baseline receives the exact label sequence $c_t, \dots, c_{t+16}$ at each decision; this sequence is also unavailable to the deployed scheduler and differs from the single derived supervision label c_t^{sup} . Table S6 states all baseline procedures and their information availability.

Table S6: Definitions of the baselines. All constrained baselines use the same candidate action set, feasibility check, channel warm-up model, and six-time-step minimum dwell-time constraint as PD-PPO unless stated otherwise.

Baseline	Decision rule	Selection and tie break	Information availability
Fixed-schedule baseline	Apply one fixed channel subset.	Evaluate all six subsets on calibration/validation windows; minimize the macro-averaged normalized forecast loss, then mean forecast loss.	Feasible baseline; no condition labels at decision time.
Fixed-priority baseline	Assign linearly decreasing scores by channel index at every time step.	Deterministic projector order.	Online and non-learned.
Round-robin baseline	Give the highest scores to a cyclic group of two channels.	The time step and channel index define a fixed cyclic schedule; projection resolves conflicts.	Online and non-learned.
AoI-priority baseline	Score each channel by clipped time since its last ready sampling opportunity.	Project the descending score order.	Online and non-learned.
Random-feasible baseline	Draw independent Gaussian channel scores at each time step.	One random stream per simulation seed, seeded by simulation seed plus 100; the same 512-score stream restarts at each event-balanced test-window reset; project the sampled ranking.	Online stochastic baseline; no within-seed Monte Carlo averaging.
Double DQN training configuration	Evaluate six Q-values and greedily select the largest currently feasible action during test evaluation.	Double DQN target with a feasible-action mask; deterministic lowest-index tie break.	Online learned baseline; no condition labels at decision time.
Warning-rule baseline	Choose the largest particle, flux, or thermal warning signal when it is at least 0.5; otherwise choose the calm action.	On calibration/validation windows, separately minimize calm or event-type loss for each warning-signal case; break ties by mean forecast loss.	Uses simulated warning signals; no condition labels at decision time.
Privileged one-step look-ahead reference using next-step test targets	Simulate every candidate for one time step and select the smallest next-step forecast loss on the test partition under the frozen primary forecaster.	Lower switching fraction, then lower action index.	Uses next-step test targets unavailable to the deployed scheduler; the objective omits minimum dwell-time, warm-up, and observation-age consequences.
Privileged true-label baseline	Choose the optional channel selected on the calibration/validation partition using the exact label sequence $c_t, \dots, c_{t+16}$.	Use the first active event type in the future-label interval; break calibration/validation ties by mean forecast loss.	Uses condition labels unavailable to the deployed scheduler.
Full-observation baseline	Request all six channels at every time step.	No selection.	Deliberately removes the constraints; unconstrained benchmark.

S7. Simulation validation and additional results

Table S7: Sensitivity to evaluation coverage in the test partition. A positive paired difference means that the fixed-schedule baseline has higher loss than PD-PPO.

Evaluation scope	Time steps per seed	Mean forecast loss (PD-PPO / fixed baseline)	Seeds with lower loss; mean paired difference [95% bootstrap CI]	Macro-averaged loss (PD-PPO / fixed baseline)	Seeds with lower loss; mean paired difference [95% bootstrap CI]
Event-balanced evaluation windows	4,096	1.4073 / 1.5744	22 of 22; +0.1671 [+0.1231, +0.2164]	0.9602 / 1.0440	22 of 22; +0.0838 [+0.0708, +0.0967]
Continuous full-partition evaluation	5,242	0.6644 / 0.7992	22 of 22; +0.1347 [+0.0995, +0.1749]	0.4318 / 0.5154	22 of 22; +0.0836 [+0.0684, +0.0996]

Each interval is a 95% bootstrap confidence interval for the paired difference. The event-balanced evaluation uses eight separately reset 512-step rollouts. Continuous full-partition evaluation uses the same final policy checkpoint and the fixed channel subset selected on the calibration/validation partition once over [64750, 69992). The last eight time steps are excluded because an eight-step target is unavailable; no policy, subset, or normalization factor is reselected.

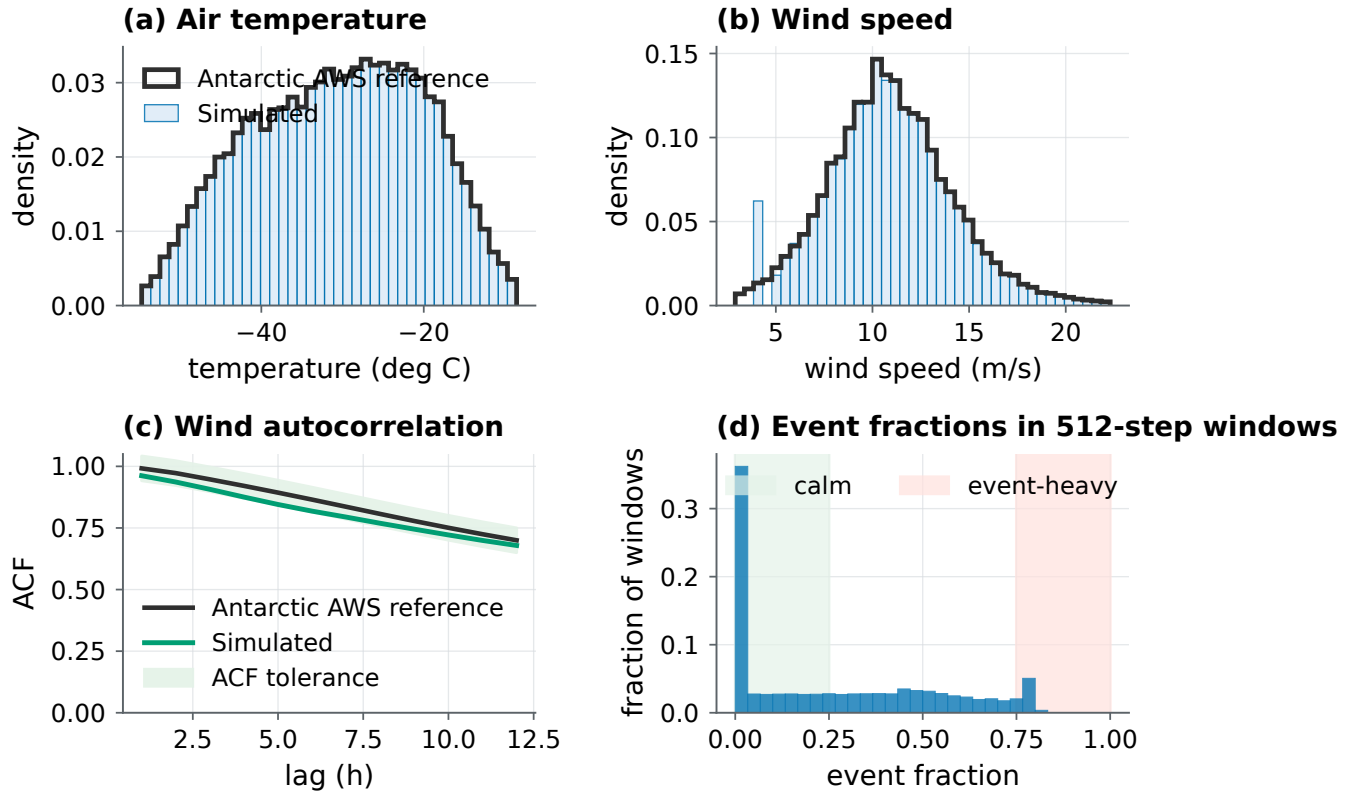


Figure S3: Simulation-validation diagnostics covering agreement between simulated and Antarctic-reference distributions, the wind-speed autocorrelation function (ACF), and event fractions.

Table S8: Alternative-forecaster analysis on unchanged observation sequences. Positive macro-averaged loss differences favor PD-PPO.

Fixed-schedule baseline	Seeds with lower loss	Mean paired difference [95% bootstrap CI]
Primary-forecaster fixed-schedule baseline	22 of 22	+0.1668 [+0.1340, +0.2044]
Ridge-forecaster fixed-schedule baseline	21 of 22	+0.1339 [+0.1103, +0.1563]

Table S9: Fixed channel subset selected for each seed on calibration/validation data.

Seed	Fixed channel subset	Selection loss	Seed	Fixed channel subset	Selection loss
117	Core + FC4 mass flux	0.907	129	Core + FC4 mass flux	0.892
118	Core + FC4 mass flux	0.928	130	Core + snow-surface temperature	0.820
119	Core + snow-surface temperature	0.804	131	Core + FC4 mass flux	0.873
120	Core + FC4 mass flux	0.901	132	Core + FC4 mass flux	0.915
121	Core + FC4 mass flux	0.859	133	Core + snow-surface temperature	0.825
122	Core + snow-surface temperature	0.844	134	Core + snow-surface temperature	0.940
123	Core + FC4 mass flux	0.905	135	Core + laser disdrometer	0.930
124	Core + snow-surface temperature	0.847	136	Core + FC4 mass flux	0.892
125	Core + snow-surface temperature	0.926	137	Core + laser disdrometer	0.880
126	Core + FC4 mass flux	0.887	138	Core + FC4 mass flux	0.888
127	Core + FC4 mass flux	0.904	139	Core + snow-surface temperature	0.821
128	Core + FC4 mass flux	0.933	140	Core + snow-surface temperature	0.905

Values are macro-averaged normalized forecast losses on the calibration/validation partition; lower is better. Compact subset labels pair the meteorological core channel with the FC4 snow mass-flux channel, infrared snow-surface-temperature channel, or laser disdrometer channel. The selected fixed-schedule baseline is evaluated unchanged on the test windows.

Table S10: Information availability for scheduler inputs, policy training, and evaluation.

Quantity	Source	Available to PD-PPO at decision time	Use in the protocol
Sample-and-hold values	Valid measurements with last-value carry-forward	Yes	Scheduler and frozen-primary-forecaster input
Observation mask and channel sampling age	Channel operating state and valid-observation log	Yes	Scheduler input; also used by rule-based baselines
Recent-history features	Last 16 sample-and-hold values and observation masks	Yes	Scheduler input; no condition label required
Previously executed channel subset and channel operating state	Scheduler execution record	Yes	Scheduler input, feasible-action check, switching cost, and execution history
Remaining required activation time r_t	Scheduler execution record	No	Environment execution guard; omitted from the active actor and critic inputs
Simulated warning signals	Noisy lead signals from the simulated blowing-snow event process	Yes	Scheduler input under the simulation assumption
Current condition label c_t	Simulator ground truth at time t	No	Training-loss weighting; offline evaluation weights and grouping
Future-condition label c_t^{sup}	supervision First non-calm label in $[t, t + 16]$, or calm if none occurs	No	Behavior cloning pretraining, continuing advantage-weighted behavior cloning, and auxiliary future-condition classification during policy training
Label sequence $c_t, \dots, c_{t+16}$	Exact current and future simulator labels	No	Privileged decision-time input used only by the true-label baseline
Future target values	Simulated ground-truth time series	No	Training-only information for forecaster training and the reward; evaluation-only information for forecast loss
Normalization condition-to-action mapping, and fixed schedule	factors, Calibration/validation partition map-	No	Training-loss normalization, supervised-target construction, fixed-schedule selection, and metric normalization

The warning diagnostics use receiver operating characteristic area under the curve (ROC AUC), precision–recall area under the curve (PR AUC), and the macro-averaged F1 score.

Table S11: Baselines, privileged references, and training-configuration comparisons.

Baseline or variant	Role in the protocol	Interpretation
Fixed-schedule baseline	Select one feasible fixed channel subset before test evaluation.	Primary fixed-schedule baseline.
AoI-priority, round-robin, and random-feasible baselines	Use feasible cycling, AoI priority, or random rules.	Non-learned dynamic baselines.
Double DQN training configuration	Learn with the same scheduler input, forecast-loss reward, candidate action set, and feasibility check.	Training-configuration comparison; only the complete PD-PPO configuration uses behavior cloning pretraining, continuing advantage-weighted behavior cloning, and an auxiliary future-condition classifier.
Privileged one-step look-ahead reference using next-step test targets	Choose the currently best feasible action using next-step forecast loss on the test partition.	Uses next-step target values unavailable during deployment and does not isolate sequential decision value.
Warning-rule baseline	Map simulated warning signals to channel subsets selected on calibration/validation data.	Memoryless baseline using the supplied simulated warning signals.
Privileged true-label baseline	Map the exact label sequence $c_t, \dots, c_{t+16}$ to channel subsets selected on calibration/validation data.	Uses condition labels that are unavailable during deployment.
Scalar-objective PPO configurations	Retrain the same PPO architecture and constraints with AoI or a bounded diagonal uncertainty proxy as the reward.	Matched PPO training configurations with different scalar objectives; objective-dependent advantages also change the PPO surrogate, critic targets, and behavior cloning weights.

Table S12: Descriptive aggregate over all 24 reported seeds, including the two pilot/model-selection seeds. Positive differences favor the complete PD-PPO training configuration. These values are retained for transparency and are not the post-selection inferential analysis.

Comparison and scope	Endpoint	Seeds favoring PD-PPO	Mean paired difference [95% bootstrap CI]
Event-balanced evaluation windows: fixed-schedule baseline	Mean forecast loss	24 of 24	+0.1580 [+0.1161, +0.2052]
Event-balanced evaluation windows: fixed-schedule baseline	Macro-averaged loss	24 of 24	+0.0801 [+0.0673, +0.0931]
Continuous full partition: fixed-schedule baseline	Mean forecast loss	24 of 24	+0.1247 [+0.0898, +0.1641]
Continuous full partition: fixed-schedule baseline	Macro-averaged loss	24 of 24	+0.0793 [+0.0642, +0.0951]
Warning-rule baseline	Macro-averaged loss	11 of 24	+0.0012 [-0.0043, +0.0076]
Privileged true-label baseline	Macro-averaged loss	12 of 24	+0.0018 [-0.0039, +0.0081]
Double DQN training configuration	Macro-averaged loss	24 of 24	+0.0697 [+0.0539, +0.0855]
AoI-objective PPO configuration	Macro-averaged loss	13 of 24	+0.0010 [-0.0039, +0.0058]
Uncertainty-objective PPO configuration	Macro-averaged loss	12 of 24	+0.0008 [-0.0036, +0.0046]
Ridge-forecaster fixed-schedule baseline	Macro-averaged loss	23 of 24	+0.1334 [+0.1112, +0.1544]

The objective rows are matched PPO training configurations with different scalar objectives, not isolated scalar-reward ablations. The Double DQN row compares complete training configurations. The warning-rule and privileged true-label confidence intervals include zero.

Table S13: Information content of the simulated warning signals over the continuous test partitions of the 22 post-selection evaluation seeds.

Diagnostic	Value
Thresholded four-class warning rule	Accuracy 0.9094; macro-averaged F1 score 0.9156
Mutual information with current condition label	1.4711 bits
Particle warning	ROC AUC 0.9971; PR AUC 0.9855
Flux warning	ROC AUC 0.9988; PR AUC 0.9881
Thermal warning	ROC AUC 0.9971; PR AUC 0.9832
Event-onset warning lead	1,262 of 1,262 onsets detected in the preceding 16 steps; median 9 steps

Diagnostics are computed directly from hash-verified frozen truth CSVs; they do not execute a policy. The rule selects the largest event warning when it is at least 0.5 and calm otherwise.

Table S14: Constraint and execution diagnostics for the reported $q = 1$ candidate-action instance.

Diagnostic	Value
Candidate / feasible action count	6 / 6 at every step
$\Pr(\mathcal{I}_t < 6)$	0
Maximum steady demand / limit	0.74 / 0.75; zero exclusions
Maximum startup-peak demand / limit	0.92 / 0.95; zero exclusions
Minimum dwell time / maximum warm-up	6 / 2 time steps
Warm-up interruptions and $\Pr(W_t > 0)$	0 and 0, structurally
Mean normalized Hamming distance per transition	0.00369 over concatenated event-balanced evaluation windows
Nominal $\lambda_{\text{sw}} S_t$ from that diagnostic	3.69×10^{-6} ; not a logged training-reward contribution
Training-time proposal override rate	Not recorded; counterfactual proposals are absent from logs